Tutorial 10

Exercise 1: Matchings

Let (V, E) be an undirected graph. A matching is a subset $M \subseteq E$ of edges such that for all vertices $v \in V$, at most one edge in M is incident to v (i.e., (V, M) does not contain a path containing more than two vertices). A maximum matching is a matching M such that we have $|M| \geq |M'|$ for every matching M'.

Furthermore, we say that (V, E) is bipartite if $V = L \cup R$ such that

- L and R are disjoint ($L \cap R = \emptyset$) and
- all edges in E go between L and R ($(u, v) \in E$ implies $\{u, v\} \cap L \neq \emptyset$ and $\{u, v\} \cap R \neq \emptyset$).

1. Describe a task that can be solved by modelling it as an maximum matching problem in a bipartite undirected graph. Bonus points if it involves pizza.
2. Show how to reduce the maximum matching problem in bipartite undirected graphs to the maximum flow problem.
3. Show how to express the maximum matching problem in bipartite undirected graphs as a linear program.

Exercise 2: Disjoint paths

Let (V, E) be a directed graph with two dedicated vertices $s \neq t \in V$. A path from s to t is a sequence $v_0 \cdots v_n$ of vertices such that $v_0 = s$, $(v_i, v_{i+1}) \in E$ for all $0 \leq i < n$, and $v_n = t$. We say that the path contains the vertices $v_0, v_1, \ldots, v_n$ and that it contains the edges $(v_0, v_1), (v_1, v_2), \ldots, (v_{n-1}, v_n)$.

- We say that two paths from s to t are vertex disjoint if the only vertices that are contained in both paths are s and t.
- We say that two paths from s to t are edge disjoint if there is no edge that is contained in both paths.

The maximum vertex-disjoint (egde-disjoint) path problem asks, given a directed graph (V, E) and vertices $s \neq t \in V$, to compute the maximal number of vertex disjoint (edge disjoint) paths from s to t.

1. Describe a problem that can be solved by modelling it as an maximum vertex disjoint path problem in a directed graph.
2. Describe a problem that can be solved by modelling it as an maximum edge disjoint path problem in a directed graph.
3. Show how to reduce the maximum vertex disjoint path problem in directed graphs to the maximum flow problem.
4. Show how to express the maximum vertex disjoint path problem in directed graphs as a linear program.
5. Show how to reduce the maximum edge disjoint path problem in directed graphs to the maximum flow problem.
6. Show how to express the maximum edge disjoint path problem in directed graphs as a linear program.

Exercise 3: The mimimum-cost flow problem

In the minimum-cost flow problem, each edge (u, v) in a flow network comes with a cost-coefficient $a(u, v)$ and a throughput $d \in \mathbb{R}$. One is interested in determining a flow f of value d (if one exists) that minimizes the cost of the flow, which is obtained by summing up the cost of f for each edge (u, v), defined as $a(u, v) \cdot f(u, v)$.

1. Define the minimum-cost flow problem formally.
2. Show how to express the minimum-cost flow problem as a linear program.